Department of Computer Science and Engineering Bangladesh University of Business and Technology (BUBT)

CSE 498A: Literature Review Records

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Capstone Project Title	Multi-Protocol Label Switching (MPLS) in a Service Provider Network.
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Aspects	Paper #1 Automated Root Cause Analysis of Network Failures in IP/MPLS Network Using Machine Learning and Case-Based Reasoning. [Published Year: 2025, Publisher: IEEE]
Problem Statement	The paper addresses the inherent complexity of managing large-scale IP/MPLS networks, where chain failures a single fault cascading across multiple devices—are challenging to diagnose manually. Traditional manual troubleshooting is time-consuming, prone to human error, and inefficient due to the volume and heterogeneity of event logs (e.g., Syslog, SNMP Traps) and the multi-layered, meshed topology of such networks.
Key Contributions & Objectives/Goal	1.An automated fault identification system integrating multiple modules for rapid root cause analysis. 2. Real-time event processing using Apache Kafka for scalable data handling. 3. A supervised learning model for classifying inconsistent event messages. 4. Actionable operational insights via dashboards and notifications.

Methodology/Theory/Framework	The system uses a dual-layer architecture: 1.Log Analysis Layer: Standardizes logs via template generation (wildcard replacement) and classifies events using TF-IDF and SVM. 2.) Operation and Maintenance Layer: Performs event analysis, node chain lookup, node testing (via SSH), and case-based reasoning for root cause identification.
Software Tools/Setup Details	The system uses Apache Kafka for message brokering, KSQL for stream processing, Confluent Platform for deployment, Paramiko for SSH-based node testing, and NLTK for text preprocessing. The dashboard and notification service use modern messaging platforms like Line.

Test/Experiment Analysis	Performance was evaluated using a server with an Intel Xeon CPU and 16 GB RAM. Metrics included precision, recall, F1-score, and processing time per service. A sample of 5,000 event messages was used to measure efficiency.
Test Data/Dataset Source	Over 7 million event messages were collected from routers, switches, DWDM devices, and servers via NMS between January 2023 and August 2024. These were categorized into 20 event types.
Final Result (Assessment Criteria Wise)	The SVM model achieved the highest F1-score of 0.969. The system processed non-fault messages in ~0.7 seconds, while fault-critical messages requiring node tests took ~41 seconds. The dashboard and notification system provided real-time alerts and root cause details.
Limitations	1.The model requires retraining when new devices are added. 2. Limited to single-event processing; no inter-event relationship analysis. 3. Node chain lookup is restricted to one-level neighbors. 4. Node testing is slow due to sequential SSH commands.
Final Summary	This paper presents an automated root cause analysis system for IP/MPLS networks that combines machine learning and case-based reasoning. It uses a dual-layer architecture to standardize and classify event messages, achieving high accuracy with an SVM model. Real-time processing is enabled via Apache Kafka, and operators receive actionable insights through a dashboard and notifications. Despite its effectiveness, the system has limitations in scalability, device adaptability, and processing speed for complex tests. Future work may include parallel processing and LLM integration for enhanced decision-making.

Aspects	**Paper #2 Research on the Application of MPLS VPN Technology in Data Transmission [Published Year: 2025, Publisher: ACM]**

Problem Statement	The paper identifies the primary challenge of ensuring secure and reliable data transmission for space Telemetry, Tracking, and Command (TT&C) activities. Traditionally, this relies on self-built or leased fiber-optic lines with Optical Link Protection (OLP), which introduces operational complexities and new security risks. The problem is to find a solution that can quickly establish a secure, private network for mobile TT&C equipment in complex environments without the high cost and inflexibility of physical dedicated lines.
Key Contributions & **Objectives/Goal**	The key contributions and objectives of the article are: 1. To propose and validate MPLS VPN technology as a solution for building virtual private networks over public infrastructure for mobile TT&C data transmission. 2. To experimentally demonstrate the feasibility of establishing MPLS VPNs both within a single Autonomous System (AS) and across different ASs (i.e., different network operators). 3.To analyze and compare network performance (specifically latency and routing efficiency) before and after implementing MPLS VPN.

	3. To highlight the advantages of MPLS VPN over traditional OLP dedicated lines, including improved security, scalability, and simpler operation and maintenance.
Methodology/Theory/Framework	The methodology is a combination of theoretical explanation and experimental simulation. 1.Theory: The paper first explains the working principles of MPLS (Layer 2.5 protocol using labels for forwarding), VPN (creating a private network over a shared one), and their combination, MPLS VPN (using label switching to create secure tunnels). 2.Framework: The solution is framed around implementing MPLS VPN in two key scenarios_ Intra-AS: Building a VPN within a single network operator's domain. Inter-AS: Establishing a VPN across different network operators' domains using MP-EBGP (Multiprotocol External Border Gateway Protocol) over LDP LSPs (Label Distribution Protocol Label Switched Paths).
Software Tools/Setup Details	The experimental simulation was performed using Huawei eNSP (Enterprise Network Simulation Platform), a network modeling tool for simulating Huawei routers and switches.

Test/Experiment Analysis (How is the experimental analysis performed? And, what parameters are used during experimental analysis?)	The experimental analysis was performed by: 1. Building network topologies in eNSP that simulate both intra-AS and inter AS environments. 2. Configuring devices as CE (Customer Edge), PE (Provider Edge), P (Provider), and ASBR (Autonomous System Border Router) with appropriate routing protocols (OSPF, BGP). 3. Comparing key performance indicators before and after MPLS VPN implementation. The primary parameter used for analysis was network latency (minimum, maximum, and average). 3. Verifying information isolation by simulating multiple data streams and checking routing tables to ensure they remained separate and inaccessible to unauthorized VPNs.
Test Data/Dataset Source (What dataset is used?)	The paper did not use an external dataset. The test data was generated synthetically within the simulation environment. The "data" consisted of network traffic (e.g., ping commands) between simulated hosts (PC1, PC2) across the constructed MPLS VPN tunnels. The IP address schemes (e.g., 192.168.0.0/8, 10.1.0.0/16) were configured by the authors as part of the simulation setup.
Final Result (Assessment Criteria Wise)	The experiments successfully demonstrated that: 1.Latency Improvement: After enabling MPLS VPN, the network latency was reduced. The routing tables were simplified by hiding intermediate backbone routes, leading to more efficient data forwarding. 2.Successful Cross-Domain Setup: The establishment of MPLS VPN channels across different ASs (operators) was achieved, proving the feasibility of inter operator private lines.

	3.Security and Isolation: The technology effectively isolated different user VPNs, preventing unauthorized access to routing information and data streams. 4.Operational Efficiency: The solution was found to be simpler to configure and maintain than traditional networks, as adding a new service only requires configuration on the PE devices connected to the customer, without altering the core network topology.
Limitations	The authors did not explicitly state any major limitations of their study. However, based on the content, an implicit limitation is that the research is entirely based on simulation results from Huawei eNSP. The performance and feasibility were not validated on a real-world, physical network, which may have different characteristics and challenges.

Final Summar	This paper investigates using MPLS VPN to create secure, virtual private networks over public infrastructure for mobile TT&C data transmission. It addresses the limitations of traditional dedicated fiber lines by proposing a more flexible and scalable solution. Through simulations in Huawei eNSP, the authors successfully built MPLS VPNs both within and across different network operators' domains. The results demonstrated that MPLS VPN reduces network latency, simplifies routing, and provides secure data isolation. The study concludes that MPLS VPN is a feasible, available, and reliable technology for rapidly deploying TT&C networks in complex environments, offering advantages in security and maintainability over traditional methods.

Aspects	**Paper #3 MPLSNetwork Actions: Technological Overview and P4- Based Implementation on a High-Speed Switching ASIC.** **[Published Year: 2025, Publisher: IEEE]**
Problem Statement	The paper addresses the limited extensibility of the traditional MPLS protocol, which relies on a small number of Special Purpose Labels (SPLs) for new functions. It highlights that alternative solutions like IPv6 Extension Headers are poorly supported in hardware due to their unbounded flexibility, which prevents high-speed processing. The need is for a new, hardware feasible framework to extend MPLS with advanced functions without sacrificing performance.
Key Contributions & **Objectives/Goal**	The key contributions of the article are: 1.A comprehensive technological overview of the MNA framework, its concepts, and identified use cases. 2. A comparison between MNA and IPv6 Extension Headers, arguing MNA's better deployability.

	3. The first hardware implementation of MNA (P4-MNA) on the Intel Tofino™ 2 ASIC, achieving 400 Gb/s line rate. 4. Evaluation of P4-MNA's scalability, performance, and implementation of two use cases: AMM for packet loss measurement and network slicing for bandwidth reservation. 5. Identification of header stacking constraints and a proposal to signal maximum NAS sizes to integrate hardware with smaller RLD. 6. Analysis of challenges with mutable data in ISD and an outlook on Post-Stack Data (PSD).
Methodology/Theory/Framework	The methodology is based on the IETF's proposed MNA framework. This framework repurposes MPLS label stack entries to create Network Action Sub-stacks (NAS). A NAS contains a base SPL indicator, followed by opcodes that define specific actions and ancillary data fields for parameters. The framework uses scopes (Hop-by-Hop, Select, Ingress-to-Egress) to control which nodes along a path execute the actions.
Software Tools/Setup Details	The implementation uses the P4 language to program the data plane of an Intel Tofino 2 switching ASIC. The control plane is written in Rust using the rbfrt library. For testing, the authors used P4TG, a custom high-speed traffic generator, and developed auxiliary tools like a Wireshark dissector for MNA.
Test/Experiment Analysis	Experimental analysis was conducted in a testbed with Tofino 2 switches. Performance was measured by transmitting traffic with varying numbers of network actions at 400 Gb/s and assessing Round-Trip Time (RTT) and packet loss. For the use cases, the AMM action was validated by comparing measured packet loss against configured loss rates, while the network slicing action was tested by verifying that bandwidth reservations were enforced during congestion.
Test Data/Dataset Source	The test data was synthetically generated and not from a public dataset. The authors used their P4TG traffic generator to create Constant Bit Rate (CBR) traffic with custom MPLS stacks that included the specific network actions under evaluation.
Final Result (Assessment Criteria Wise)	The P4-MNA implementation successfully achieved line-rate processing at 400 Gb/s with zero packet loss. The processing delay introduced by network actions was a negligible 13.3 nanoseconds per hop. Both implemented use cases worked precisely: the AMM action correctly measured configured packet loss, and the network slicing action guaranteed bandwidth by dropping only non-conforming traffic.

Limitations	1. The In-Stack Data (ISD) approach is inefficient for mutable data. 2. Only about **30% of the bits in a NAS** can be modified during transit due to compatibility requirements. 3. The implementation is limited by the hardware's **Readable Label Depth (RLD)**. 4. The evaluation tested **only 11 network actions**, even though the implementation can support **up to 32 actions**.
Final Summary	This paper provides a comprehensive overview and the first hardware implementation of the emerging MPLS Network Actions framework. It demonstrates that MNA is feasible for high-speed programmable hardware, achieving 400 Gb/s line rate with minimal performance impact. The implementation successfully validated two practical use cases: performance measurement and network slicing. However, the work also identifies key constraints, such as the high RLD requirements and the inefficiency of ISD for mutable data, proposing the signaling of maximum NAS sizes as a solution for resource-constrained devices. The analysis concludes that while MNA is more hardware-friendly than IPv6 Extension Headers, careful design is needed to ensure broad deployability across different hardware platforms.

Aspects	**Paper #4 Independent Time Transfer in 5G Technology Over IP Core Networks Using PTN Overlay and Comparison With GPS/GNSS-Based Synchronization** **[Published Year: 2025, Publisher: IEEE]**
Problem Statement	The paper addresses the critical dependency of 5G networks on GPS/GNSS for time synchronization, which is vulnerable to spoofing, jamming, and signal degradation in urban environments. This reliance creates a single point of failure that threatens network reliability for essential 5G services like URLLC and mMTC. Additionally, alternative network-based solutions like ITU-T G.8275.x require expensive upgrades to PTP-aware infrastructure, presenting substantial cost barriers for operators with legacy equipment.
Key Contributions & **Objectives/Goal**	The key contributions of the article are: 1.A comprehensive comparison of the Precision Time Network (PTN) overlay with GPS/GNSS and PTP synchronization.

	2. The design and implementation of a PTN-based architecture using a Dynamic Synchronous Transfer Mode (DTM) overlay for independent time transfer over an IP core network. 3. Extensive real-world performance measurements from a multi-city testbed in Turkey over a two-year period. 4. Demonstration of PTN's ability to achieve sub-100 ns synchronization accuracy, sub-500 ms recovery times, and enhanced resilience to network impairments. 5.Analysis of the cost-effectiveness and security advantages of leveraging existing IP infrastructure compared to GNSS or full PTP-aware network upgrades.
Methodology/Theory/Framework	The solution is based on a PTN architecture that uses a DTM overlay on top of an existing IP/MPLS network. A centralized Time of Day (ToD) reference in Kayseri distributes timing information via virtual DTM trunks to nodes across the country. This method leverages DTM's deterministic, circuit-switched characteristics to minimize jitter and packet delay variation, enabling precise time transfer even over networks without full timing support. The system was validated against local GNSS references at the edge nodes.
Software Tools/Setup Details	2.Dedicated synchronization nodes were used to build the PTN. 4. A Time Interval Counter (TIC) validated measurements by comparing PTN PPS signals with a GNSS-disciplined reference. 3. TimeLab and Stable32 handled data recording and post-processing for time-error and stability analysis.
Test/Experiment Analysis	Experimental analysis was conducted over a two-year period across a multi city testbed in Türkiye. Key performance parameters included mean time error (nanoseconds), jitter (time error variation), and recovery time (milliseconds) following network disruptions. The system's resilience was tested during an actual MPLS network outage, with statistical analysis using confidence intervals to validate reliability under real-world conditions.
Test Data/Dataset Source	The test data originated from the authors' operational testbed rather than public repositories. The dataset comprises timing measurements collected over three-week periods from nodes deployed across multiple Turkish cities including Istanbul, Ankara, and Bursa, replicating authentic urban traffic patterns and network conditions.

Final Result (Assessment Criteria Wise)	The PTN overlay consistently achieved sub-100 ns synchronization accuracy with an average time error of 92.3 ns across all nodes. Jitter remained low at 25.6 ns average, and the system demonstrated exceptional resilience by

	recovering from a major MPLS outage in under 500 ms. These results significantly outperform GNSS systems that typically require manual intervention taking seconds to minutes for recovery from jamming or spoofing incidents.
Limitations	1.The PTN testbed's data capture system could store only **three weeks of data,** restricting long-term stability analysis. 2. The initial implementation required an **external PPS-to-PTP conversion mechanism,** though a newer integrated version is now commercially available. 3. PTN is resilient to **GNSS-related threats,** but remains vulnerable to **network-based attacks (DDoS).** 4. Additional security measures such as **encryption and authentication** are necessary to mitigate these vulnerabilities.
Final Summary	This paper presents and validates a GNSS-independent time synchronization method for 5G core networks using a Precision Time Network with a DTM overlay. Through a large-scale, real-world testbed in Turkey, the study demonstrates that PTN can reliably achieve sub-100 nanosecond accuracy, outperforming GPS/GNSS in terms of security against jamming and spoofing. The solution leverages existing IP infrastructure, offering a cost-effective alternative to expensive full network PTP upgrades. Its rapid recovery from network failures and deterministic performance make it a highly resilient and practical option for meeting the stringent synchronization demands of future 5G and beyond applications.

Aspects	**Paper #5 Stack Management for MPLS Network Actions: Integration of Nodes with Limited Hardware Capabilities [Published Year: 2025, Publisher: ACM]**
Problem Statement	The paper identifies that the MPLS Network Actions (MNA) framework increases the required readable label depth (RLD) in routers due to the addition of multiple Label Stack Entries (LSEs) for network actions. This leads to inefficient use of hardware resources because routers must parse irrelevant LSEs in the "in-between stack" to locate the hop-by-hop (HBH) scoped Network Action Sub-stack (NAS). This wasteful parsing increases the RLD requirement, making it difficult for routers with limited hardware

	capabilities to support MNA at line rate.
Key Contributions & Objectives/Goal	The key contributions of the article are: 1. A hardware analysis of an MNA implementation identifying the root cause of high RLD requirements. 2. Proposal of the HBH preservation mechanism to reduce RLD by restructuring the MPLS stack during forwarding.

	3. Introduction of a novel stack management network action to enable the HBH preservation mechanism. 4. A solution for integrating the mechanism in networks with MNA-incapable nodes. 5. A P4-based implementation on the Intel Tofino™ 2 ASIC demonstrating feasibility at 400 Gb/s. 6. Analysis of the impact on RLD, packet overhead, and ECMP.
Methodology/Theory/Framework	The solution involves the HBH preservation mechanism, which ensures that the HBH-scoped NAS remains just below the top-of-stack label by moving forwarding labels from below the NAS to the top after popping the top label. This eliminates the need to parse the in-between stack. A stack management network action is introduced to control this label movement, with parameters to handle backward compatibility in mixed-capability networks.
Software Tools/Setup Details	The proposed mechanism is implemented in P4 and deployed on the Intel Tofino™ 2 switching ASIC. The implementation extends the existing P4-MNA codebase from prior work, and the source code is publicly available on GitHub.
Test/Experiment Analysis	Experimental analysis is performed by verifying the HBH preservation mechanism in a network topology that includes both MNA-capable and MNA-incapable nodes. The implementation is tested for functional correctness and performance at a line rate of 400 Gb/s per port. Parameters such as RLD, number of consecutive MNA-incapable nodes, and packet header overhead are considered.
Test Data/Dataset Source	No external dataset is used. The evaluation is based on a custom network scenario designed to demonstrate the mechanism's operation and backward compatibility, as illustrated in the paper's figures.
Final Result (Assessment Criteria Wise)	The HBH preservation mechanism reduces the required minimum RLD to 36 LSEs, compared to the original implementation which required more due to the in-between stack. The mechanism operates at line rate, avoids unnecessary packet copies, and maintains ECMP consistency. The stack management network action adds only 4 bytes of overhead per action.

Limitations	The authors note that the HBH preservation mechanism is not supported on legacy MPLS hardware, which lacks support for network actions. Additionally, the minimum RLD of 36 LSEs may still be too high for some resource constrained devices, though this can be mitigated by reducing the maximum NAS size.
Final Summary	The paper addresses the high RLD requirement in MNA-enabled networks by proposing a stack management mechanism that restructures the MPLS stack during forwarding. The HBH preservation mechanism eliminates the need for parsing irrelevant LSEs, thereby reducing the RLD and improving resource efficiency. A novel stack management network action enables this mechanism and ensures compatibility with MNA-incapable nodes. The solution is implemented in P4 on programmable hardware and verified at high data rates. While legacy hardware limitations remain, the approach significantly enhances the practicality of deploying MNA in heterogeneous networks.